

SSC JE MAINS CIVIL EXAM STRATEGY

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SSC JE PREVIOUS YEAR EXAM ANALYSIS :-

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022	2023	2024	2025
Vacancies	1407	844	1718	1242	918	277	1506	743	541	2139	1145	1252	1312

NO. OF Application form-

YEAR	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022	2023	2024	2025
NO. OF FORM	102145	165661	264945	308852	339379	569930	813622	806078	660629	641952	562813	4,83,557	4,65,439

Year	Total Candidates appeared in Exam	Candidates qualified in paper-I (Civil)	Candidates qualified in paper-II (Civil)	Final selected Candidates (Civil)
2012	102,145	5025	2680	1407
2013	165,661	4809	1980	844
2014	264,945	8852	3567	1718
2015	308,852	6788	3291	1242
2016	339,379	5892	2765	918
2017	569,930	4433	1280	256
2018	377,133	8697	3800	1506
2019	225,723	4750	2532	743
2020	284,047	3826	1294	541
2022	309,145	15605	2139	2139
2023	310,877	10154	1145	1145
2024	2,70,791	11765	1252	1252

Note: NO SSC JE RECRUITMENT IN 2021

SSC JE Paper 1 Cut Off (Out of 200):-

Year	UR	OBC	SC	ST	OH	HH	EWS
2012	62.25	52.5	47.75	43.5	30	30	-
2013	78	70.5	66.25	63.5	60	40	-
2014	93.75	82	75.75	70	69	40	-
2015	103.75	91.25	88	87.75	78	30	-
2016	100	92.5	84.5	58.5	72.5	40	-
2017	117	110.75	101.75	105	91.5	61.75	-
2018	127.4	122.91	107.61	107.01	97.45	61.61	118.99
2019	123.52	115.93	101.7	102.61	92.24	55.73	112.28
2020	120.02	114.21	99.15	99.15	78.83	48.86	108.14
2022	110.57	107.99	86.36	86.32	80.28	40	89.08
2023	108.16	106.5	89.36	87.33	84.62	56.45	98.91
2024	122.92	116.82	101.85	97.83	84.29	40.07	109.20

SSC JE Paper 2 Cut Off (Out of 500):-

Year	UR	OBC	SC	ST	OH	HH	EWS
2012	146.25	106.5	109.75	83.5	70	75	-
2013	161	139.5	128.25	119.5	119	72	-
2014	229.75	182	155.75	149	148	70	-
2015	234.75	153.25	138	137.75	128	70	-
2016	220.5	186	164	163.75	139.5	87.5	-
2017	244.75	244.75	220.75	228	231.25	152	-
2018	250.49	209.38	193.68	201.54	162.01	132.68	-
2019	315.55	265.07	235.51	243	169.02	122.79	270.44
2020	257.84	234.28	201.59	188.11	150.23	137.52	229.58
2022	323.4	310.53	290.01	296.86	244.69	186.65	314.32

After 2022 SSC JE PAPER 2 CUTOFF & FINAL CUTOFF ARE SAME.BECAUSE SSC TAKE NO. OF CANDIDATES = VACANCY FOR DV NOW.

SSC JE FINAL CUTOFF :-

Year	UR	OBC	SC	ST	EWS
2012	184-276.75	190-258.5	168-238.75	144.25-237.25	-
2013	199.75-280.75	226-291.5	213.25-245.5	205-232.25	-
2014	285-353	287.75-334.5	265.25-307.75	250-392.75	-
2015	235.5-284.75	236.25-276.25	213.5-289.75	236.25-267.75	-
2016	238.25-264.75	236-252.25	205.25-225	228.25-245	-
2017	289.75-301.75	283.5-292.0	262.75-272.25	279.0-291.0	-
2018	295-275	280-250	270-235	270-245	295-265
2019	362.02-319.14	349.50-305.23	320.17-272.13	318.56-267.61	348.99-304.64
2020	306.76-264.13	280.83-261.3	258.75-227.34	248.38-216.32	297.14-252.26
2022	347.37-323.40	345.43-310.53	322.01-290.01	328.88-296.86	339.41-314.32
2023	325.96-346.73	319.90-339.94	293.90-311.65	299.96-313.86	319.94-338.17
2024	425.93-375.92	394.21-369.87	378.56-349.55	378.38-348.18	393.68-368.80

SSC JE MAINS SYLLABUS and EXAM PATTERN:

Papers	Mode of Exam	Subjects	No. of Questions/Marks	Time	Negative Marks
PAPER-II	CBT	Civil Engineering	100/300	2 Hours	1/3 rd = 1

NOTE - IN SSC JE MAINS EXAM CALCULATOR, IS CODE DATA During online exam.
Final merit list will be combined of both papers' marks.

MAINS SYLLABUS

Building Materials: Physical and Chemical properties, classification, standard tests, uses and manufacture/quarrying of materials e.g. building stones, silicate based materials, cement (Portland), asbestos products, timber and wood based products, laminates, bituminous materials, paints, varnishes.

Estimating, Costing and Valuation: estimate, glossary of technical terms, analysis of rates, methods and unit of measurement, Items of work – earthwork, Brick work (Modular & Traditional bricks), RCC work, Shuttering, Timber work, Painting, Flooring, Plastering. Boundary wall, Brick building, Water Tank, Septic tank, Bar bending schedule, Centre line method, Mid-section formula, Trapezoidal formula, Simpson's rule. Cost estimate of Septic tank, flexible pavements, Tube well, isolates and combined footings, Steel Truss, Piles and pile-caps. Valuation – Value and cost, scrap value, salvage value, assessed value, sinking fund, depreciation and obsolescence, methods of valuation.

Surveying : Principles of surveying, measurement of distance, chain surveying, working of prismatic compass, compass traversing, bearings, local attraction, plane table surveying, theodolite traversing, adjustment of theodolite, Levelling, Definition of terms used in levelling, contouring, curvature and refraction corrections, temporary and permanent adjustments of dumpy level, methods of contouring, uses of contour map, tachometric survey, curve setting, earth work calculation, advanced surveying equipment.

Soil Mechanics: Origin of soil, phase diagram, Definitions-void ratio, porosity, degree of saturation, water content, specific gravity of soil grains, unit weights, density index and interrelationship of different parameters, Grain size distribution curves and their uses. Index properties of soils, Atterberg's limits, IS soil classification and plasticity chart. Permeability of soil, coefficient of permeability, determination of coefficient of permeability, Unconfined and confined aquifers, effective stress, quick sand, consolidation of soils, Principles of consolidation, degree of consolidation, pre-consolidation pressure, normally consolidated soil, e-log p curve, computation of ultimate settlement. Shear strength of soils, direct shear test, Vane shear test, Triaxial test. Soil compaction, Laboratory compaction test, Maximum dry density and optimum moisture content, earth pressure theories, active and passive earth pressures, Bearing capacity of soils, plate load test, standard penetration test.

Hydraulics: Fluid properties, hydrostatics, measurements of flow, Bernoulli's theorem and its application, flow through pipes, flow in open channels, weirs, flumes, spillways, pumps and turbines.

Irrigation Engineering: Definition, necessity, benefits, 2II effects of irrigation, types and methods of irrigation, Hydrology – Measurement of rainfall, run off coefficient, rain gauge, losses from precipitation – evaporation, infiltration, etc. Water requirement of crops, duty, delta and base period, Kharif and Rabi Crops, Command area, Time factor, Crop ratio, Overlap allowance, Irrigation efficiencies. Different type of canals, types of canal irrigation, loss of water in canals. Canal lining –types and advantages. Shallow and deep to wells, yield from a well. Weir and barrage, Failure of weirs and permeable foundation, Slit and Scour, Kennedy's theory of critical velocity. Lacey's theory of uniform flow. Definition of flood, causes and effects, methods of flood control, water

logging, preventive measure. Land reclamation, Characteristics of affecting fertility of soils, purposes, methods, description of land and reclamation processes. Major irrigation projects in India.

Transportation Engineering: Highway Engineering – cross sectional elements, geometric design, types of pavements, pavement materials – aggregates and bitumen, different tests, Design of flexible and rigid pavements – Water Bound Macadam (WBM) and Wet Mix Macadam (WMM), Gravel Road, Bituminous construction, Rigid pavement joint, pavement maintenance, Highway drainage, Railway Engineering- Components of permanent way – sleepers, ballast, fixtures and fastening, track geometry, points and crossings, track junction, stations and yards. Traffic Engineering – Different traffic survey, speed-flow-density and their interrelationships, intersections and interchanges, traffic signals, traffic operation, traffic signs and markings, road safety.

Environmental Engineering: Quality of water, source of water supply, purification of water, distribution of water, need of sanitation, sewerage systems, circular sewer, oval sewer, sewer appurtenances, sewage treatments. Surface water drainage. Solid waste management – types, effects, engineered management system. Air pollution – pollutants, causes, effects, control. Noise pollution cause, health effects, control.

Theory of structures: Elasticity constants, types of beams – determinate and indeterminate, bending moment and shear force diagrams of simply supported, cantilever and over hanging beams. Moment of area and moment of inertia for rectangular & circular sections, bending moment and shear stress for tee, channel and compound sections, chimneys, dams and retaining walls, eccentric loads, slope deflection of simply supported and cantilever beams, critical load and columns, Torsion of circular section.

Concrete Technology: Properties, Advantages and uses of concrete, cement aggregates, importance of water quality, water cement ratio, workability, mix design, storage, batching, mixing, placement, compaction, finishing and curing of concrete, quality control of concrete, hot weather and cold weather concreting, repair and maintenance of concrete structures.

RCC Design: RCC beams-flexural strength, shear strength, bond strength, design of singly reinforced and double reinforced beams, cantilever beams. T-beams, lintels. One way and two way slabs, isolated footings. Reinforced brick works, columns, staircases, retaining wall, water tanks (RCC design questions may be based on both Limit State and Working Stress methods).

Steel Design: Steel design and construction of steel columns, beams roof trusses plate girders.

Subject-Wise Analysis of SSC JE CIVIL PREVIOUS YEAR'S PAPERS

PRE

Year	BMC	SOM	Estimation	Survey	RCC	Steel	FM	WRE	Environment	Transportation	Soil
2004	15	5	0	7	6	8	11	0	10	0	12
2005	12	10	0	5	9	8	11	0	9	0	11
2007	14	5	0	5	15	10	8	0	10	0	8
2008	17	9	0	7	11	7	7	0	7	0	10
2009	18	9	0	7	9	8	7	0	6	0	10
2010	22	12	0	8	16	12	11	0	8	0	11
2011	17	15	6	7	12	10	8	4	6	7	8
2011 CWC	12	7	0	3	8	5	3	3	3	3	4
2012	16	22	8	11	12	5	9	2	5	2	5
2013	20	25	5	12	13	3	9	2	4	2	5
2014 (M)	29	12	8	10	8	11	11	2	2	3	4
2014 (E)	28	9	5	11	11	12	9	3	2	2	4
2015	28	10	6	10	12	8	9	1	4	1	7
1/3/2017 (M)	29	7	7	10	17	10	10	2	2	2	4
1/3/2017 (E)	31	12	7	10	11	9	10	2	2	2	4
2/3/2017 (M)	22	23	9	10	11	4	10	2	2	2	5
2/3/2017 (E)	27	13	10	10	12	6	10	2	2	2	6
3/3/2017 (M)	28	28	7	7	3	2	9	3	2	4	4
3/3/2017 (E)	22	17	9	10	11	8	11	1	5	2	4
4/3/2017 (M)	27	14	7	10	10	7	10	3	2	4	4
4/3/2017 (E)	29	13	7	10	10	8	10	3	4	2	3
23/9/2019(M)	15	6	9	10	9	6	8	10	8	7	7
23/9/2019(E)	14	8	9	8	13	5	7	9	7	8	10
25/9/2019(M)	11	4	11	10	14	6	8	9	9	8	9
28/10/2020(E)	14	2	12	9	10	7	10	8	8	8	8
29/10/2020(M)	13	4	12	9	10	7	6	11	8	10	8
30/10/2020(M)	17	4	12	9	9	6	9	10	7	7	7
30/10/2020(E)	12	5	13	9	9	4	8	8	7	9	11
11/12/2020(E)	9	4	10	10	11	5	8	7	8	9	8
23/3/2021 (M)	10	4	10	10	11	7	11	10	8	9	9
23/3/2021 (E)	11	5	9	12	9	7	10	7	10	9	9
14/11/2022(M)	15	8	9	10	6	8	9	9	8	12	7
15/11/2022(M)	15	6	10	10	10	5	9	10	9	8	7
15/11/2022(E)	15	6	10	10	6	9	9	8	10	9	7
16/11/2022(M)	14	5	9	11	9	6	9	10	9	10	8
9/10/2023	12	6	7	12	10	5	11	12	7	6	12
10/10/2023 (M)	8	4	8	13	14	5	13	9	8	7	11
10/10/2023 (E)	11	5	4	12	12	5	9	11	11	11	9
11/10/2023	14	6	10	11	13	2	13	8	8	9	6
5/6/2024	16	4	5	9	12	1	3+5	8+1	14	11	10
6/6/2024 (M)	14	6	6	8	8	2	9+2	9+4	11	12+1	8
6/6/2024 (E)	16	5+2	6	8	13	-	7+2+2	7+3	9	11+1	8
3/12/2025 (M)	17	3+2	9	9	12		8+3+2	2+7	11	3+1	11
4/12/2025 (M)	15	5+1	11	9	10	2	4+4+5	5+4	10	2+2	11
5/12/2025(M)	18	3+3	9	10	11	-	10+1+2	8+3	9	4	9
5/12/2025(E)	22	7+3	4	4	9	10	4+1	2+11	6	10+1	8
6/12/2025 (M)	17	7+2	9	9	8	-	8+3+3	5+4	10	6	11
6/12/2025 (E)	18	4+4	4	4	10	14	3+2	11+2	6	8	10
13/12/2025 (M)	21	5	6	2	7	15	8	8+4	5	7+3	9

- SOM- SOM+ STRUCTURE ANALYSIS
- WRE- IRRIGATION + HYDROLOGY
- FM- FM+HYDRAULIC MACHINE + OPEN CHANNEL FLOW
- TRANSPORTATION -HIGHWAY +RAILWAY +AIRPORT

SSC JE MAINS CIVIL SUBJECT-WISE ANALYSIS

YEAR	BMC	ESTIMATION	SURVEY	SOIL	HYDRAULICS	IRRIGATION + HYDROLOGY	Transportation	ENVIRONMENT	SOM+ Structure Analysis	CONCRETE TECHNOLOGY	RCC	STEEL
2004	70		30	30	60		-	30	60	50	60	30
2005	30		40	50	30			40	60	20	45	45
2007	30		40	40	40			30	60	30	60	30
2008	30		30	55	30	15		30	60	30	60	30
2009	30		30	60	30			30	42	48	60	30
2010	30		30	60	30			30	45	45	60	30
2011	15	15	15	45	18	27	15	10	45	30	110	15
2012	40		40	20	20	20	20	40	65		55	40
2013	45	30	15	30	15	30	15	15	60	15	60	30
2014	30	30	30	15	30	15	15	15	30	50	40	60
2015	45		30	30	30	15	15	40	30	15	60	50
2016	15		30	75	30	15	30	50	40		50	25
2017	10	10	30	75	30	50	30	15	25	20	40	25
2018	10	15	15	80	15	75	30	15	20	20	40	25
2019	25	20	35	50	30	20	30	30	30	30	30	30
2020	30	15	15	95	30	45	30	15	20		40	25
2022	30	30	30	45	30	0	30	45	35	30	30	25
2023	36	30	30	33	30	15+15	21	18	18+0	9	24	24
2024	45	27	27	24	9+9+9	24+3	24+6	24	18+0	15	18	18

From 2023 objective pattern starts. Each question - 3 marks

- HYDRAULICS- FM+HM+OCF
- TRANSPORTATION - HIGHWAY+RAILWAY+AIRPORT

EDUQUITY PATTERN :-

SSC JE PRE 7 SHIFTS ANALYSIS:

Year	BMC	SOM	Estimation	Survey	RCC	Steel	FM	WRE	Environment	Transportation	Soil
3/12/2025 (M)	17	3+2	9	9	12		8+3+2	2+7	11	3+1	11
4/12/2025 (M)	15	5+1	11	9	10	2	4+4+5	5+4	10	2+2	11
5/12/2025(M)	18	3+3	9	10	11	-	10+1+2	8+3	9	4	9
5/12/2025(E)	22	7+3	4	4	9	10	4+1	2+11	6	10+1	8
6/12/2025 (M)	17	7+2	9	9	8	-	8+3+3	5+4	10	6	11
6/12/2025 (E)	18	4+4	4	4	10	14	3+2	11+2	6	8	10
13/12/2025 (M)	21	5	6	2	7	15	8	8+4	5	7+3	9

Subject Name	Merged Topics from All Shifts
Building Materials & Concrete Technology	Cement: Setting time, Soundness, Heat of hydration (Calorimeter), Types. Bricks: Sizes, Frog, Strength, Efflorescence. Concrete: W/C ratio, Workability (Slump test), Bulking of sand, Admixtures, Curing. Timber: Seasoning, Structure, Defects, Plywood. Stones: Classification, Hardness, Quarrying.
Fluid Mechanics	Properties: Viscosity, Surface tension, Capillarity. Equations: Bernoulli's, Continuity, Darcy-Weisbach (Head loss). OCF: Efficient sections, Froude No, Hydraulic jump. Machinery: Pumps (Priming, NPSH, Slip), Turbines (Specific speed, Efficiency), Venturimeter.
Soil Mechanics	Basics: Void ratio, Porosity, Moisture content, Atterberg limits. Classification: IS & USCS systems. Permeability: Darcy's Law, Anisotropy. Strength & Pressure: Shear strength (Mohr-Coulomb), Active/Passive Earth pressure. Tests: Proctor Compaction, SPT (N-value), Sieve analysis, Foundations.
Surveying	Linear: Metric chains (links), Tapes, Corrections (Pull, Scale). Angular: Compass, Bearings (WCB/QB), Local attraction, Theodolite adjustments. Levelling: HI & RL methods, Curvature/Refraction corrections, Bubble tube sensitivity. Methods: Plane Table (Resection/Intersection), Contouring, EDM, GPS basics.
RCC Design	Principles: LSM & WSM, Exposure classes. Sections: Singly/Doubly/Under-reinforced, Moment of resistance. Components: Beams (Shear steel), Slabs (1-way/2-way), Columns (Effective length), Retaining walls, Lintel. Details: Bond stress, Development length, Hooks.
Environmental Engineering	Water: Quality standards (pH, Hardness, Fluoride), Treatment (Filtration, Disinfection/Chlorine, Alum). Sewage: BOD/COD, Sewer self-cleansing velocity. Pollution: Air (Smog, NOx), Noise levels (CPCB), EIA, Solid waste management (Incineration, Composting).
Estimating, Costing & Valuation	Estimates: Detailed, Plinth area, Centreline method. Valuation: Scrap/Salvage value, Depreciation (Straight line), Sinking fund, Capitalized value. Analysis: Rate analysis, Units of measurement, Plastering deductions, Standard rent.
Irrigation & Hydrology	Water Req: Duty, Delta, Base period, Efficiency. Hydrology: Hydrographs (Unit/Base flow), Rain gauges, Double mass curve. Infiltration: ϕ -index, Transpiration. Canals: Lining, Efficient sections, Muskingum method (Flood routing).
Structural Analysis & SOM	Stress/Strain: Hooke's Law, Stress-strain curve, Elastic constants, Poisson's ratio. Bending: SFD & BMD, Section modulus, Pure bending. Components: Moment of Inertia, Deflection, Columns (Euler's formula), Conjugate beam method, Izod test.
Transportation Engineering	Highway: Geometric design (SSD, Camber, Super-elevation, Gradient), Transition curves. Traffic: Volume, Density, Speed, Mean headway. Railway: Rails, Sleepers (density), Gauges, Cant.
Steel Design	Rivet diameter, Slenderness ratio, Plastic hinge, Gantry girder, Plate girder (stiffeners).

NUMERICALS: Soil Mechanics , Hydrology Max Weightage

Exam Date	No. Of Q	Numerical Topics
03 Dec 2025	22	RL calculation, Tape corrections, Compaction, SSD, Built-up area, Runoff rate, Volumetric strain
04 Dec 2025	20	Dry volume, Tiles calculation, Seepage (Flow net), BOD removal, Jam density, FSI, Discharge in channel
05 Dec 2025M	15	Water demand, Chlorine dosage, Tachy multiplying constant, Sinking fund, Neutral axis depth, Discharge velocity
05 Dec 2025A	12	Void ratio, Capitalized value, Built-up area, Duty/Delta, Neutral axis calculation
06 Dec 2025M	16	Map scale, Local attraction, Porosity, Dry unit weight, Centerline length, Direct runoff
06 Dec 2025A	11	SSD calculation, Transition length, Depreciation, Plasticity index, Stress in steel
13 Dec 2025	19	Heat of hydration (Calorimeter), W/C ratio, Sieve analysis, Head loss (Darcy), RL numericals

PAPER LEVEL EASY – AS COMPARE TO OLD TCS PAPERS